

Start with any line: random w_1, w_2, b

Pick one training point at random

On the correct side?

no

Move the line towards it:
 $w \leftarrow w + \eta (y - \hat{y}) x$

yes: do nothing

no

Stop?

yes

Keep the final w_1, w_2, b

Two stopping rules

1. a fixed number of loops (for example 1,000)
2. convergence: no training point is misclassified